

FMDPT-7000 Smart Differential Pressure Transmitter

The FMDPT-7000 is a high-performance Smart Differential Pressure Transmitter engineered for demanding industrial flow, level, and differential pressure measurement. Built on a rugged, all-welded stainless steel sensing platform, it delivers dependable accuracy across an ultra-wide measurement span — 20 mbar to 100 bar — with a 100:1 turn-down ratio, making it a versatile choice for process plants, refineries, and power stations.

General Features

- Ultra-wide range: 20 mbar to 100 bar, 7 selectable spans, with 100:1 turn-down.
- Best-in-class accuracy: $\pm 0.05\%$ URL (optional) / $\pm 0.075\%$ URL, with $\pm 0.2\%$ SPAN / 10-year long-term stability.
- HART communication with 4-20 mA DC output; optional integrated Bluetooth (iB) for wireless configuration.
- Global hazardous-area certifications: ATEX, IECEx, CSA, PCEC, NEPSI, EAC.
- IP67 rated; optional 3-valve or 5-valve integrated manifold assembly.



Key Specifications

Measurement	Differential Pressure
Range	20 mbar to 100 bar (7 ranges)
Accuracy	$\pm 0.05\%$ (opt.) / $\pm 0.075\%$ URL
Turn-Down	100:1
Output	4-20 mA HART, optional Bluetooth
Certification	ATEX, IECEx, CSA, PCEC, NEPSI, EAC
IP Rating	IP67

Industries & Applications

- Oil & Gas — wellhead, midstream pipeline, and downstream refinery DP measurement.
- Power Generation — boiler feedwater, steam drum level, and heat exchanger fouling detection.
- Chemical & Petrochemical — corrosive fluid flow and level with Hastelloy / Tantalum wetted parts.
- Water & Wastewater — filter differential, pump protection, and tank level monitoring.
- Marine & Offshore — DNV, LR, ABS, BV, KR, ClassNK, RS certified for vessel and FPSO applications.

General Specifications — FMDPT-7000

Parameter	Specification
Measurement Type	Differential Pressure
Nominal Range	20 mbar to 100 bar (7 selectable ranges)
Turn-Down Ratio	100:1
Accuracy (Linear)	±0.05% URL (optional) / ±0.075% URL
Long-Term Stability	±0.2% Span / 10 years
Output Signal	4-20 mA DC with HART; optional integrated Bluetooth (/B)
Supply Voltage	18.3-44 V DC (HART); 18.3-30 V DC (intrinsically safe HART); 13 V DC option available
Power Consumption	500 mW max @ 24 V DC, 20.8 mA
Load Resistance	0-1476 Ohm (operating); 250-800 Ohm (HART communication)
Damping (Electronic)	0-100 s configurable; sensor module damping ≤0.2 s
Response Time	≤100 ms
IP Rating	IP67
Weight	Approx. 4 kg (excluding mounting bracket and process accessories)

Measurement Range Specifications

Nominal Range	Min. Range	Lower (LRL)	Upper (URL)	Static Press.	HP/LP Overload
20 mbar	2 mbar	-20 mbar	20 mbar	420 bar	250 bar
60 mbar	2 mbar	-60 mbar	60 mbar	420 bar	250 bar
0.4 bar	4 mbar	-0.4 bar	0.4 bar	420 bar	250 bar
2.5 bar	25 mbar	-2.5 bar	2.5 bar	420 bar	250 bar
10 bar	0.1 bar	-10 bar	10 bar	420 bar	250 bar
30 bar	0.3 bar	-30 bar	30 bar	420 bar	250 bar
100 bar	1 bar	-3 MPa	100 bar	420 bar	200 bar

Note: LRV/URV may be set within measurement range limits; it is recommended to choose a range ratio with the minimum possible value.

Accuracy — Linear Output

Condition	Range	Accuracy
$TD \leq 5$	20 mbar, 60 mbar	0.075%
$TD \leq 5$	0.4, 2.5, 10, 30 bar	0.05% (optional) / 0.075%
$TD > 5$	20 mbar, 60 mbar	$\pm(0.001 + 0.0148 \times TD)\%$
$TD > 5$	0.4, 2.5, 10, 30, 100 bar	$\pm(0.0275 + 0.0095 \times TD)\%$

Note: Square-root output accuracy is 1.5× linear output accuracy. TD (Turn-Down) = URL / Current SPAN. Reference: 20°C ±5°C, calibrated from zero.

Static Pressure Effects

Range	Effect
Range ≤ 0.1 bar	$\delta \leq \pm 0.5\% \text{ F.S. / 100 bar}$
0.1 bar < Range ≤ 0.4 bar	$\delta \leq \pm 0.1\% \text{ F.S. / 100 bar}$
2.5 bar ≤ Range ≤ 10 bar	$\delta \leq \pm 0.075\% \text{ F.S. / 100 bar}$
30 bar ≤ Range ≤ 100 bar	$\delta \leq \pm 0.15\% \text{ F.S. / 100 bar}$

Environmental Conditions

Parameter	Specification
Ambient Temperature	Without LCD: -50 to +85°C; With LCD: -40 to +70°C
Storage Temperature	Without LCD: -50 to +100°C; With LCD: -40 to +85°C
Sensor Operating Temperature	-40 to +105°C
Medium Temp. (Silicone Fill)	-40 to +200°C (standard); -55 to +200°C (low-temp silicone)
Medium Temp. (Fluorocarbon)	-20 to +110°C (Fluorocarbon 1); -55 to +85°C (Fluorocarbon 2, oxygen service)
Ambient Humidity	5% RH to 100% RH @ 40°C
Voltage Effect	For supply voltage within 18.3-44 V DC, zero/span change ≤ 0.005% SPAN/V
Vibration Effects	< 0.1% Span per GB/T18271.3 / IEC 61298-3
Mounting Position Error	< 4 mbar (correctable by PV = 0 reset at installation)

EMC Performance

Test	Standard	Test Condition	Level
Radiated Interference (Housing)	GB/T 9254.1 / CISPR 32	30 MHz - 1000 MHz	Qualified
Conducted Interference (DC Port)	GB/T 9254.1 / CISPR 32	0.15 MHz - 30 MHz	Qualified
Electrostatic Discharge Immunity	GB/T 17626.2 / IEC 61000-4-2	8 kV contact / 15 kV air	B
Radiated RF/EM Field Immunity	GB/T 17626.3 / IEC 61000-4-3	10 V/m (80 MHz - 1 GHz)	A
Power Frequency Magnetic Field	GB/T 17626.8 / IEC 61000-4-8	30 A/m	B
Electrical Fast Transient / Burst	GB/T 17626.4 / IEC 61000-4-4	4 kV (5/50 ns, 100 kHz)	B
Surge Immunity	GB/T 17626.5 / IEC 61000-4-5	2 kV (L-L); 4 kV (L-G)	B
RF Conducted Disturbance Immunity	GB/T 17626.6 / IEC 61000-4-6	3 V/m (150 kHz - 80 MHz)	A

Note: Level A: normal operation within limits. Level B: temporary degradation with auto-recovery and no data loss.

Power Supply, Load & Time Index

Parameter	Specification
Power Supply Voltage	HART: 18.3-44 V DC Intrinsically safe HART: 18.3-30 V DC
Load Resistance	0-1476 Ohm (operation); 250-800 Ohm (HART communication)
Transmission Distance	< 1000 m
Power Consumption	≤ 500 mW @ 24 V DC, 20.8 mA
Damping (Electronic)	0-100 s configurable
Sensor Module Damping	≤ 0.2 s
Turn-On Time	≤ 6 s
Factory Reset Time	≤ 31 s
Response Time	≤ 100 ms

Hazardous Area Certifications

Standard	Classification Mark	Certifying Body
Ex Intrinsically Safe	Ex ia IIC T4 Ga	PCEC (China)
Ex Flameproof	Ex db IIC T6 Gb	PCEC (China)
Ex Dust Ignition-proof	Ex tb IIC T85°C Db	NEPSI (China)
ATEX Intrinsically Safe	II 1G Ex ia IIC T4 Ga	CSA / ATEX
ATEX Flameproof	II 2G Ex db IIC T6 Gb	CSA / ATEX
ATEX Dust Explosion-proof	II 2D Ex tb IIIC T80°C Db	CSA / ATEX
IECEX Intrinsically Safe	Ex ia IIC T4 Ga	IECEX / CSA
IECEX Flameproof	Ex db IIC T6 Gb	IECEX / CSA
IECEX Dust Explosion-proof	Ex tb IIIC T80°C Db	IECEX / CSA
NA Intrinsically Safe	Class I Div. 1 Grp A-D T4; Zone 0 AEx ia IIC T4 Ga	CSA
NA Flameproof	Class I Div. 1 Grp A-D T6; Zone 1 AEx db IIC T6 Gb	CSA
EAC Explosion-proof	1Ex db IIC T6 Gb; 0Ex ia IIC T4 Ga	EAC (Russia/CIS)

Note: Some certifications are in process. Contact Forbes Marshall for current certificate status and copies.

Marine & Classification Society Approvals

Society	Full Name	Certificate No.
CCS	China Classification Society	TJ23PTB00014
DNV	Det Norske Veritas	TAA00000Y1
BV	Bureau Veritas	77759/A0 BV
ABS	American Bureau of Shipping	24-2536413-PDA
LR	Lloyd's Register	LR2520052TA
KR	Korean Register	NAJ50160-AE001
NK	Nippon Kaiji Kyokai (ClassNK)	TA24376M
RS	Russian Maritime Register of Shipping	24.44.01.01398.266

Hazardous Area — Order Code (Model Code Position 1)

Code	Protection Type	Certifying Body
0	Non-hazardous area	—
1	Flameproof	PCEC (China)
2	Intrinsically Safe	PCEC (China)
3	Dust Explosion-proof	NEPSI (China)
4	Flameproof + Intrinsically Safe	PCEC / NEPSI (China)
5	Flameproof + Dust Explosion-proof + Intrinsically Safe	PCEC / NEPSI (China)
A	Flameproof	CSA
B	Intrinsically Safe	CSA
C	Flameproof + Intrinsically Safe	CSA
E	Flameproof	ATEX
F	Intrinsically Safe	ATEX
G	Flameproof + Intrinsically Safe	ATEX
J	Flameproof	IECEX
K	Intrinsically Safe	IECEX
L	Flameproof + Intrinsically Safe	IECEX
M	Explosion-proof	EAC (Russia/CIS)

Wetted Parts Material Options

Code	Diaphragm	Flange/Discharge Block	Sealing
A	316L	316	FKM
B	316L	316	Stainless steel frame PTFE
E	316L	316	Modified FKM
C	HC-276	316	FKM
D	HC-276	316	Stainless steel frame PTFE
F	HC-276	316	Modified FKM
G	Tantalum	316	FKM (extended lead time)
H	Tantalum	316	Stainless steel frame PTFE (extended lead time)
I	Tantalum	316	Modified FKM (extended lead time)

Note: FKM: -20 to 120°C. PTFE: -60 to 180°C. Modified FKM: -40 to 180°C. Tantalum diaphragm not available below 0.4 bar range.

Process Connection & Oil Filling

Code	Specification	Drain/Vent Position	Mounting
1	Female 1/4-18 NPT	Rear end of flange	Horizontal
2	Female 1/4-18 NPT	Above the side of flange	Horizontal
3	Female 1/4-18 NPT	Under the side of flange	Horizontal
4	Female 1/4-18 NPT	Side drain/vent valve	Vertical

Code	Oil Fill	Medium Temperature
S	Silicone oil	-40 to +200°C
L	Low-temperature silicone oil	-55 to +200°C (extended lead time)
D	Fluorocarbon oil 1	-20 to +110°C (extended lead time)
E	Fluorocarbon oil 2	-55 to +85°C, oxygen service (extended lead time)

Output Signal, Case & Connector Options

Option	Description
4-20 mA DC, HART	Standard 2-wire loop powered; default HART5 (HART7 selectable via /H7)
4-20 mA DC, HART + Bluetooth	Integrated BLE module for wireless configuration and monitoring via mobile app
Square Root Mode (/SQ)	Output and display in square-root format; LRL set to zero, for flow measurement
Case S/U	316L Stainless steel, dual outlet ports (F) M20x1.5 or 1/2 NPT
Case P/N	Aluminium alloy (polyester resin coated), dual outlet ports (F) M20x1.5 or 1/2 NPT

Integrated Valve Manifold & Mounting Options

The FMDPT-7000 can be factory-assembled with a 3-valve or 5-valve integrated manifold (instrument-side / process-connection-side configuration) — contract to specify complete manifold model. L-shaped pipe- or plate-mounting brackets (/G7 316 SS, /G8 Q235 carbon steel) are also available.

Option	Description
Integrated Valve Manifold (/VT)	Factory-assembled 3-valve or 5-valve manifold; contract specifies complete model
Mounting Bracket (/G7)	L-shaped bracket, 316 SS; pipe- or plate-mounted
Mounting Bracket (/G8)	L-shaped bracket, Q235 carbon steel; pipe- or plate-mounted
Process Connection Adapter (/D3)	T-shaped adapter (M), M20x1.5 with vent tube, 316 SS, x2
Process Connection Adapter (/D4)	Waist-shaped adapter, 1/2-14 NPT (F), 316 SS, x2

Ordering Information — Model Code Structure

The FMDPT-7000 model code is structured as follows. Please specify all parameters when placing an order. Items marked (Note) have extended lead times. Factory calibration range (LRL-URL) and display unit must be stated in the purchase contract.

Position	Field	Key Options
1	Hazardous Area	See Hazardous Area order-code table, page 5 (0=Non-hazardous; 1-5=China; A/B/C=CSA; E/F/G=ATEX; J/K/L=IECEX; M=EAC)
2	Output Signal	H=4-20 mA HART; B=4-20 mA HART + Bluetooth
3	Case	S=316L SS M20; U=316L SS NPT; P=Aluminium M20; N=Aluminium NPT
4	Waterproof/Explosion-proof Connector	0=None; A=Flameproof plug; 1=Waterproof; 2-5=Non-FP/FP adapters, M20-M20, NPT or G1/2
5	Display	N=Without LCD; L=With LCD
6	Range	0=20 mbar; 1=60 mbar; 2=0.4 bar; 3=2.5 bar; 4=10 bar; 5=30 bar; 6=100 bar
7	Wetted Parts	A/B/E=316L; C/D/F=HC-276; G/H/I=Tantalum (see wetted parts table)
8	Oil Filling	S=Silicone; L=Low-temp silicone; D/E=Fluorocarbon 1/2
9	Flange Bolting	1=35CrMo alloy steel; 6=316 SS (static pressure < 250 bar)
10	Flange Mounting	H=Horizontal; E=Vertical
11	Process Connection	1=Back drain; 2=Upper drain; 3=Lower drain; 4=Side drain (vertical mounting)
/Options	Certificates	/CS1=CCS; /CS2=DNV; /CS3=BV; /CS4=ABS; /CS5=LR; /CS6=KR; /CS7=NK; /CS8=RS
/Options	Special Features	/B=Bluetooth; /SQ=Square root; /ST=Damping; /J2=Gold diaphragm; /CL1=Oil-free; /WH,/WL=Fault alarm level
/Options	Other	/Q1=Verification report; /PT=ID plate; /VT=Integrated valve manifold; /QD1,/QD2=Leakage test; /HAC=High accuracy; /LD=Lithium battery
/Options	Language / Delivery	/LE,/LR=Language (English/Russian); /XM=Project-specific delivery
/Options	Warranty	/Y2=2-year; /Y3=3-year; /Y5=5-year warranty

Example: FMDPT-7000-0HS1L-1A1S6H/G1 [CAL: 0 to 50 mbar]

Dimensional Reference (mm)

Parameter	Specification
Overall Height (with LCD)	199.5 mm
Overall Height (without LCD)	144 mm
Overall Width (across port centres)	90 mm (118 mm incl. process manifold)
Depth (H & L side)	97 mm
Process Port Thread	1/4-18 NPT Female (H & L side)
Mounting Bolt Centres	54 mm x 32 mm

Note: Actual dimensions vary with process adapter and mounting bracket selection. Cable gland is supplied as an accessory to be installed by the user.

Quick Reference — Common Configurations

Worked examples for reference only; confirm every field against the full model code table before ordering.

Configuration	Order Code	Range
Standard, non-hazardous, HART, LCD	FMDPT-7000-OHS1L-1A1S6H	0-50 mbar
ATEX flameproof, HART, LCD, 3-year warranty	FMDPT-7000-EHS1L-4A1S6H/Y3	0-10 bar
CSA intrinsically safe, HART + Bluetooth, LCD	FMDPT-7000-BBS1L-4A1S6H	0-10 bar